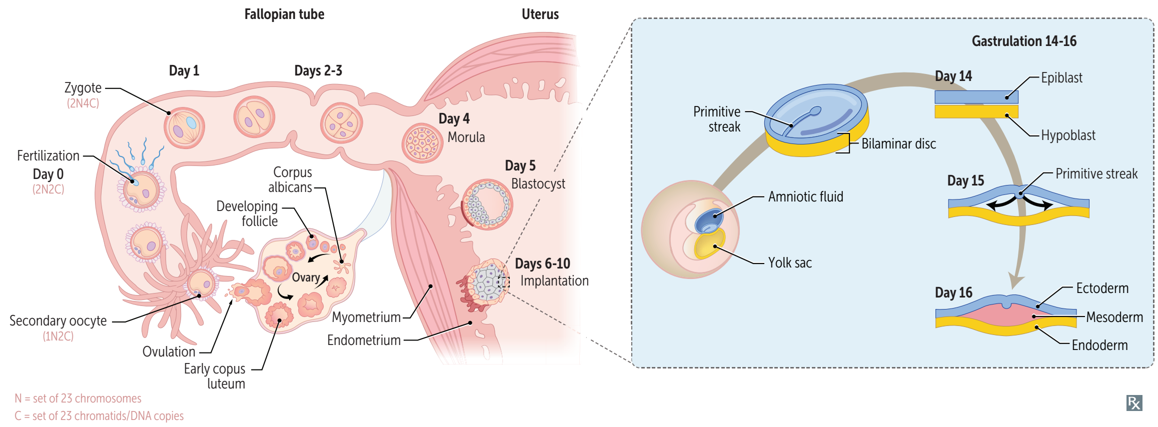
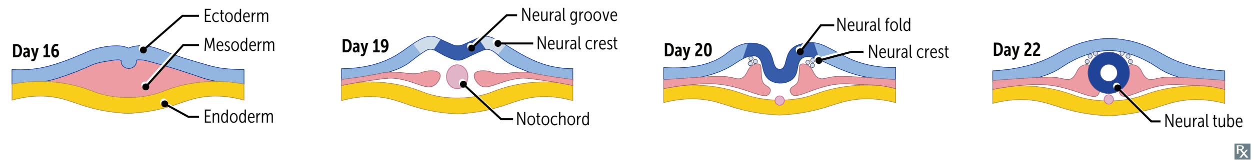
▸ REPRODUCTIVE—EMBRYOLOGY

Early embryonic development



Week 1	hCG secretion begins around the time of blastocyst implantation. Blasto**cyst** "**sticks**" on day **six**.
Week 2	Formation of **bi**laminar embryonic disc; **two** layers = epiblast, hypoblast.
Week 3	Formation of **tri**laminar embryonic disc via gastrulation (epiblast cell invagination through primitive streak); **three** layers = endoderm, mesoderm, ectoderm. Notochord arises from midline mesoderm and induces overlying ectoderm (via SHH) to become neural plate, which gives rise to neural tube via neurulation.
Week 4	Heart begins to beat (**four** chambers). Cardiac activity visible by transvaginal ultrasound. Upper and lower limb buds begin to form (**four** limbs).
Week 8	Genitalia have male/female characteristics (pronounce "gen**eight**alia").

Embryologic derivatives



Ectoderm		External/outer layer
Surface ectoderm	Epidermis; adenohypophysis (from Rathke pouch); lens of eye; epithelial linings of oral cavity, sensory organs of ear, and olfactory epithelium; anal canal below the pectinate line; parotid, sweat, mammary glands.	**Craniopharyngioma**—benign Rathke pouch tumor with cholesterol crystals, calcifications.
Neural tube	Brain (neurohypophysis, CNS neurons, oligodendrocytes, astrocytes, ependymal cells, pineal gland), retina, spinal cord.	Neuroectoderm—think CNS.
Neural crest	Enterochromaffin cells, Melanocytes, Odontoblasts, PNS ganglia (cranial, dorsal root, autonomic), Adrenal medulla, Schwann cells, Spiral membrane (aorticopulmonary septum), Endocardial cushions (also derived partially from mesoderm), Skull bones.	EMO PASSES Neural crest—think PNS and non-neural structures nearby.
Mesoderm	Muscle, bone, connective tissue, serous linings of body cavities (eg, peritoneum, pericardium, pleura), spleen (develops within foregut mesentery), cardiovascular structures, lymphatics, blood, wall of gut tube, proximal vagina, kidneys, adrenal cortex, dermis, testes, ovaries, microglia, tracheal cartilage. Notochord induces ectoderm to form neuroectoderm (neural plate); its only postnatal derivative is the nucleus pulposus of the intervertebral disc.	Middle/“meat” layer. Mesodermal defects = VACTERL association: Vertebral defects Anal atresia Cardiac defects Tracheo-Esophageal fistula Renal defects Limb defects (bone and muscle)
Endoderm	Gut tube epithelium (including anal canal above the pectinate line), most of urethra and distal vagina (derived from urogenital sinus), luminal epithelial derivatives (eg, lungs, liver, gallbladder, pancreas, eustachian tube, thymus, parathyroid, thyroid follicular and parafollicular [C] cells).	“Enternal” layer.

Teratogens Most susceptible during organogenesis in embryonic period (before week 8 of development). Before implantation, "all-or-none" effect. After week 8 (fetal period), growth and function affected.

TERATOGEN	EFFECT ON FETUS
Medications	
ACE inhibitors	Renal failure, oligohydramnios, hypocalvaria.
Alkylating agents	Multiple anomalies (eg, ear/facial abnormalities, absence of digits).
Aminoglycosides	Ototoxicity. "A mean guy hit the baby in the ear."
Antiepileptic drugs	Neural tube defects, cardiac defects, cleft palate, skeletal abnormalities (eg, phalanx/nail hypoplasia, facial dysmorphism). Most commonly due to valproate, carbamazepine, phenytoin, phenobarbital; high-dose folate supplementation recommended.
Diethylstilbestrol	Vaginal clear cell adenocarcinoma, congenital Müllerian anomalies.
Fluoroquinolones	Cartilage damage.
Folate antagonists	Neural tube defects. Most commonly due to trimethoprim, methotrexate.
Isotretinoin	Craniofacial (eg, microtia, dysmorphism), CNS, cardiac, and thymic defects. Contraception mandatory. Pronounce "isoteratinoin" for its teratogenicity.
Lithium	Ebstein anomaly.
Methimazole	Aplasia cutis congenita (congenital absence of skin, typically on scalp).
Tetracyclines	Discolored teeth, inhibited bone growth. Pronounce "teethracyclines."
Thalidomide	Limb defects (eg, phocomelia—flipperlike limbs). Pronounce "thalimbdomide."
Warfarin	Bone and cartilage deformities (stippled epiphyses, nasal and limb hypoplasia), optic nerve atrophy, cerebral hemorrhage. Use heparin during pregnancy (does not cross placenta). In war, you need strong bones to march and optics to see the enemy.
Substance use	
Alcohol	Fetal alcohol syndrome.
Cocaine	Preterm birth, low birth weight, fetal growth restriction (FGR). Cocaine → vasoconstriction.
Tobacco smoking	Preterm birth, low birth weight (leading cause in resource-rich countries), FGR, sudden infant death syndrome (SIDS), ADHD. Nicotine → vasoconstriction, CO → impaired O_2 delivery.
Other	
Iodine lack or excess	Congenital hypothyroidism.
Maternal diabetes	Caudal regression syndrome, cardiac defects (eg, transposition of great arteries, VSD), neural tube defects, macrosomia, neonatal hypoglycemia (due to islet cell hyperplasia), polycythemia, respiratory distress syndrome.
Maternal PKU	Fetal growth restriction, microcephaly, intellectual disability, congenital heart defects.
Methylmercury	Neurotoxicity. ↑ concentration in top-predator fish (eg, shark, swordfish, king mackerel, tilefish).
X-rays	Microcephaly, intellectual disability. Effects minimized by use of lead shielding.

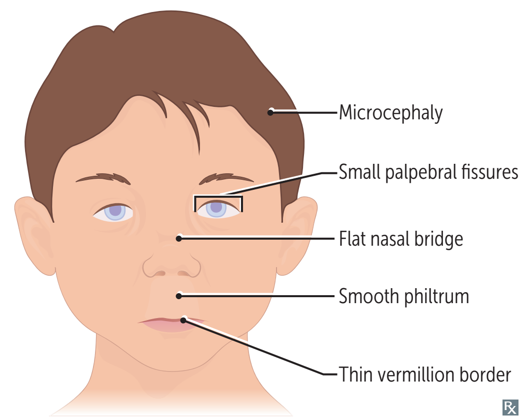
Types of errors in morphogenesis

Agenesis	Absent organ due to absent primordial tissue.
Aplasia	Absent organ despite presence of primordial tissue.
Hypoplasia	Incomplete organ development; primordial tissue present.
Disruption	2° breakdown of tissue with normal developmental potential (eg, amniotic band syndrome).
Deformation	Extrinsic mechanical distortion (eg, congenital torticollis); occurs during fetal period.
Malformation	Intrinsic developmental defect (eg, cleft lip/palate); occurs during embryonic period.
Sequence	Abnormalities result from a single 1° embryologic event (eg, oligohydramnios → Potter sequence).
Field defect	Disturbance of tissues that develop in a contiguous physical space (eg, holoprosencephaly).

Fetal alcohol syndrome

One of the leading preventable causes of intellectual disability in the US. 2° to maternal alcohol use during pregnancy. Newborns may present with developmental delay, microcephaly, facial abnormalities (eg, smooth philtrum, thin vermillion border, small palpebral fissures, flat nasal bridge), limb dislocation, heart defects. Holoprosencephaly may occur in more severe presentations. One mechanism is due to impaired migration of neuronal and glial cells.

Treatment: supportive care.



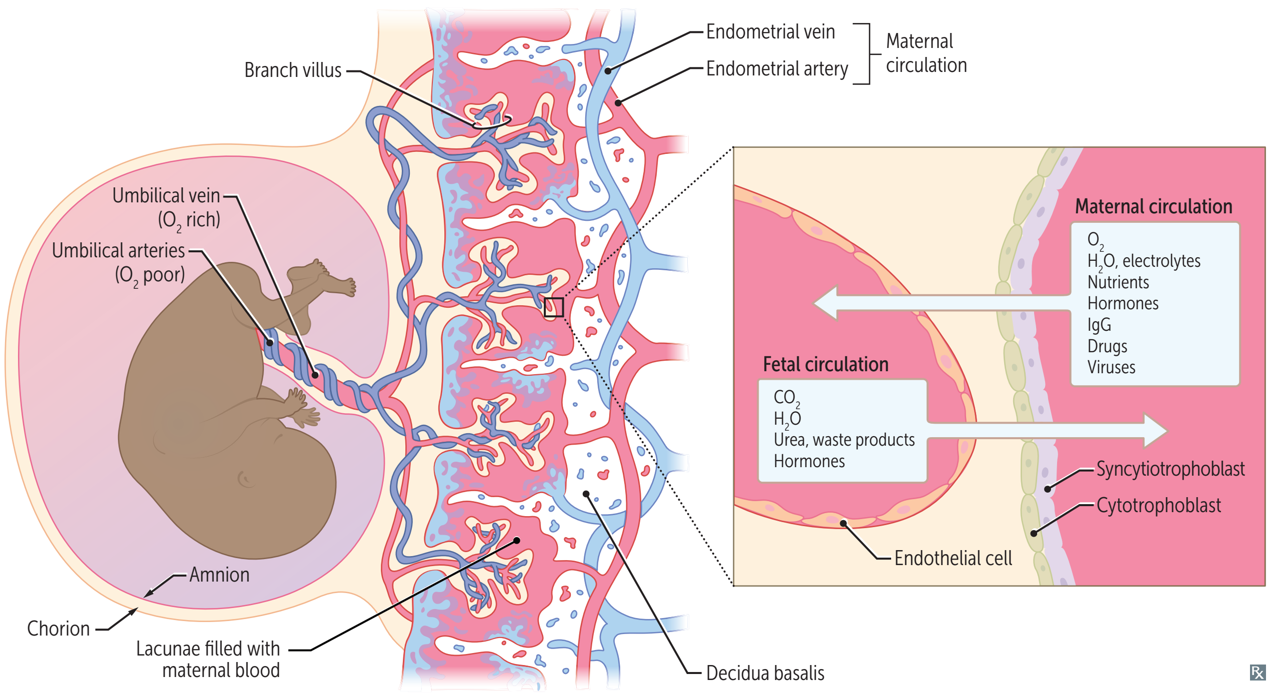
Neonatal abstinence syndrome

Complex disorder involving CNS, ANS, and GI systems. 2° to maternal substance use (most commonly opioids) during pregnancy. Newborns may present with uncoordinated sucking reflexes, irritability, high-pitched crying, tremors, tachypnea, sneezing, diarrhea, and possibly seizures.

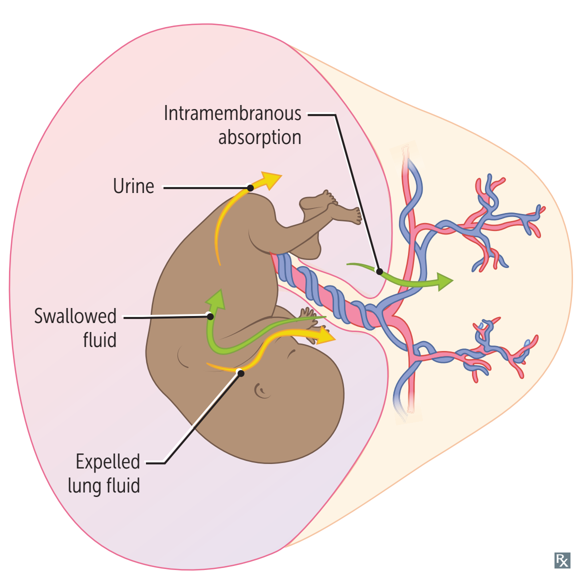
Treatment (for opioid use): methadone, morphine, buprenorphine.

Universal screening for substance use is recommended in all pregnant patients.

Placenta	1° site of nutrient and gas exchange between mother and fetus.
Fetal component	
Cytotrophoblast	Inner layer of chorionic villi; creates cells.
Syncytiotrophoblast	Outer layer of chorionic villi; synthesizes and secretes hormones, eg, hCG (structurally similar to LH; stimulates corpus luteum to secrete progesterone during first trimester). Lacks MHC I expression → ↓ chance of attack by maternal immune system.
Maternal component	
Decidua basalis	Derived from endometrium. Maternal blood in lacunae.



Amniotic fluid



Derived from fetal urine (mainly) and fetal lung liquid.

Cleared by fetal swallowing (mainly) and intramembranous absorption.

Polyhydramnios—too much amniotic fluid. May be idiopathic or associated with fetal malformations (eg, esophageal/duodenal atresia, anencephaly; both result in inability to swallow amniotic fluid), maternal diabetes, fetal anemia, multifetal gestation.

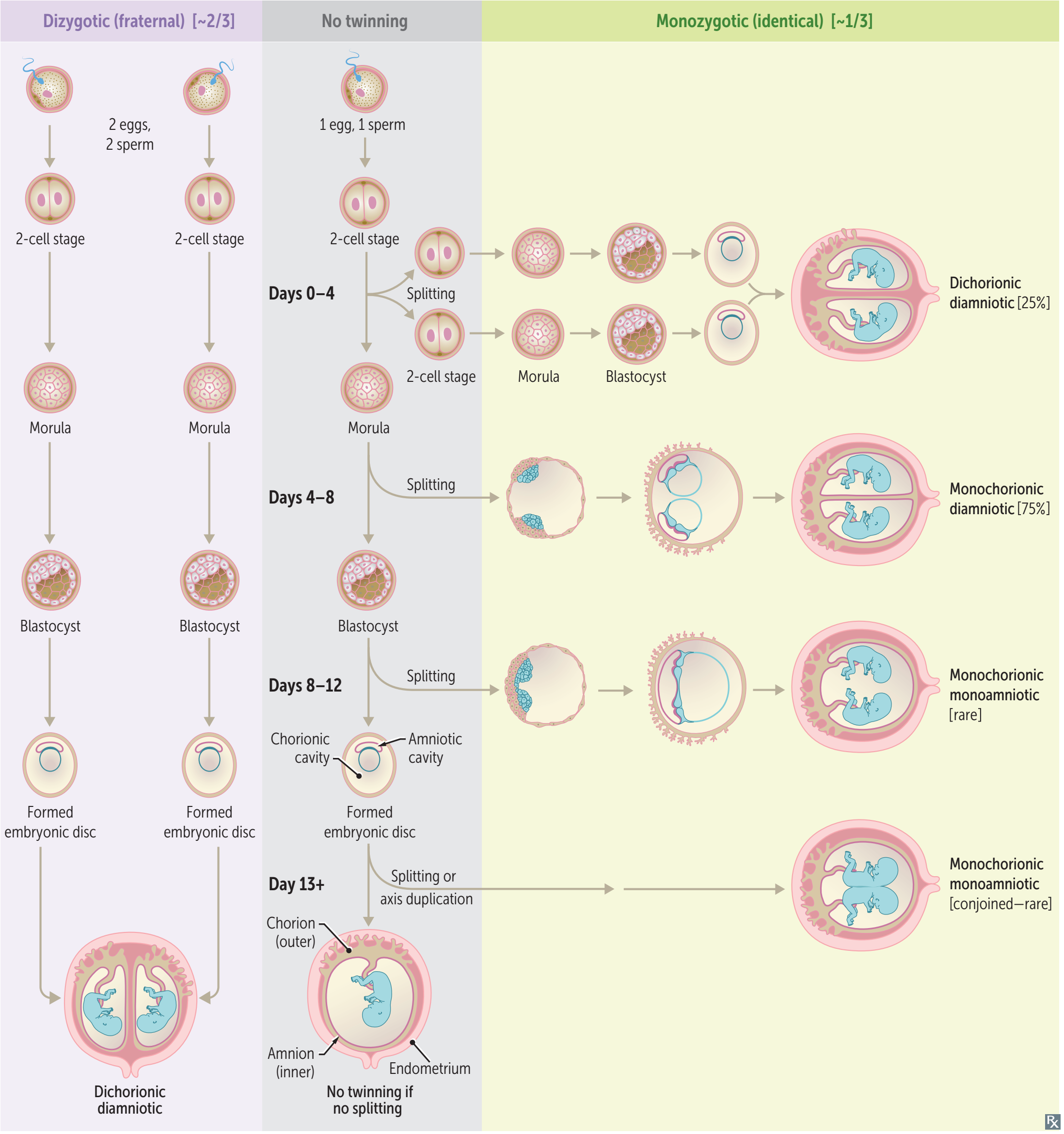
Oligohydramnios—too little amniotic fluid. Associated with placental insufficiency, bilateral renal agenesis, posterior urethral valves (in males); these result in inability to excrete urine. Profound oligohydramnios can cause Potter sequence.

Twinning

Dizygotic ("fraternal") twins arise from 2 eggs that are separately fertilized by 2 different sperm (always 2 zygotes) and will have 2 separate amniotic sacs and 2 separate placentas (chorions).

Monozygotic ("identical") twins arise from 1 fertilized egg (1 egg + 1 sperm) that splits in early pregnancy. The timing of splitting determines chorionicity (number of chorions) and amnionicity (number of amnions) (take **separate** cars or **share** a **CAB**):

- Splitting 0–4 days: **separate** chorion and amnion (di-di)
- Splitting 4–8 days: **shared** **C**horion (mo-di)
- Splitting 8–12 days: **shared** chorion and **A**mnion (mo-mo)
- Splitting 13+ days: **shared** chorion, amnion, and **B**ody (mo-mo; conjoined)



Twin-twin transfusion syndrome

Occurs in monochorionic twin gestations. Unbalanced arteriovenous anastomoses between twins in shared placenta → net blood flow from one twin to the other.

Donor twin → hypovolemia and oligohydramnios ("stuck twin" appearance).

Recipient twin → hypervolemia and polyhydramnios.

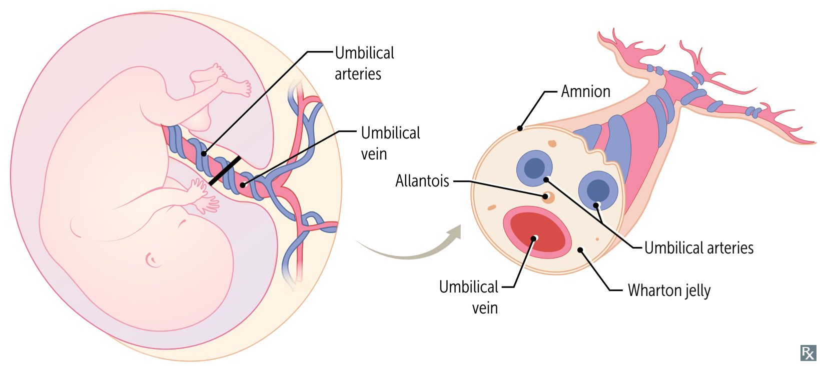
Umbilical cord

Two umbilical arteries return deoxygenated blood from fetal internal iliac arteries to placenta.

One umbilical vein supplies oxygenated blood from placenta to fetus; drains into IVC via liver or via ductus venosus.

Single umbilical artery (2-vessel cord) is associated with congenital and chromosomal anomalies.

Umbilical arteries and vein are derived from allantois.



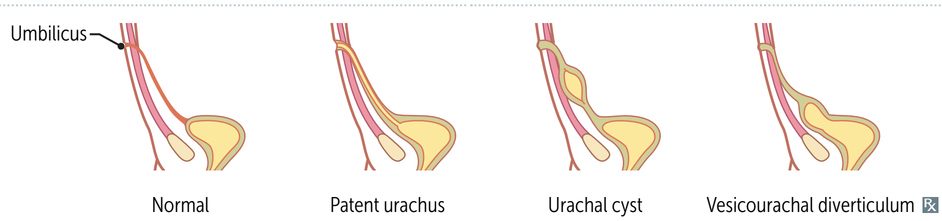
Urachus

Allantois forms from yolk sac and extends into cloaca. Intra-abdominal remnant of allantois is called the urachus, a duct between fetal bladder and umbilicus. Failure of urachus to involute can lead to anomalies that may increase risk of infection and/or malignancy (eg, adenocarcinoma) if not treated. Obliterated urachus is represented by the median umbilical ligament after birth, which is covered by median umbilical fold of the peritoneum.

Patent urachus — Total failure of urachus to obliterate → **u**rine discharge from **u**mbilicus.

Urachal cyst — Partial failure of urachus to obliterate; fluid-filled cavity lined with uroepithelium, between umbilicus and bladder. Cyst can become infected and present as painful mass below umbilicus.

Vesicourachal diverticulum — Slight failure of urachus to obliterate → outpouching of bladder.



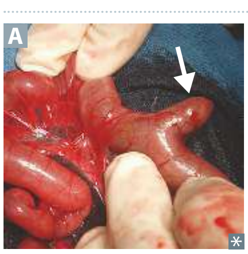
Vitelline duct

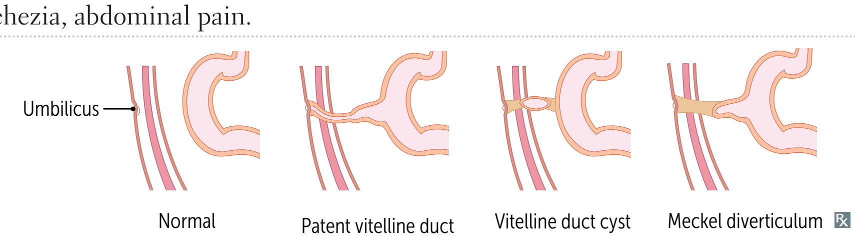
Also called omphalomesenteric duct. Connects yolk sac to midgut lumen. Obliterates during week 7 of development.

Patent vitelline duct — Total failure of vitelline duct to obliterate → meconium discharge from umbilicus.

Vitelline duct cyst — Partial failure of vitelline duct to obliterate. ↑ risk for volvulus.

Meckel diverticulum — Slight failure of vitelline duct to obliterate → outpouching of ileum (true diverticulum, arrow in A). Usually asymptomatic. May have heterotopic gastric and/or pancreatic tissue → melena, hematochezia, abdominal pain.

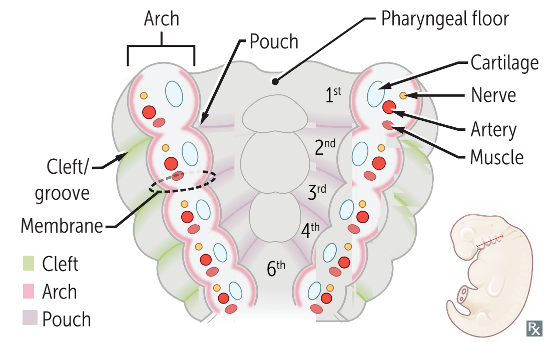




Pharyngeal apparatus

Composed of pharyngeal (branchial) clefts, arches, pouches.
Pharyngeal clefts—derived from ectoderm. Also called pharyngeal grooves.
Pharyngeal arches—derived from mesoderm (muscles, arteries) and neural crest (bones, cartilage).
Pharyngeal pouches—derived from endoderm.

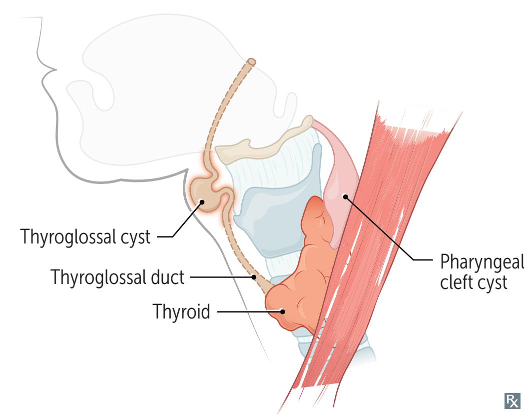
CAP covers outside to inside:
Clefts = ectoderm
Arches = mesoderm + neural crest
Pouches = endoderm



Pharyngeal cleft derivatives

1st cleft develops into external auditory meatus.
2nd through 4th clefts form temporary cervical sinuses, which are obliterated by proliferation of 2nd arch mesenchyme.

Pharyngeal cleft cyst—persistent cervical sinus; presents as lateral neck mass anterior to sternocleidomastoid muscle that does not move with swallowing (vs thyroglossal duct cyst).



Pharyngeal pouch derivatives

Ear, **tonsils**, **bottom-to-top**: 1 (**ear**), 2 (**tonsils**), 3 dorsal (**bottom** = **inferior** parathyroids), 3 ventral (**to** = **thymus**), 4 (**top** = **superior** parathyroids).

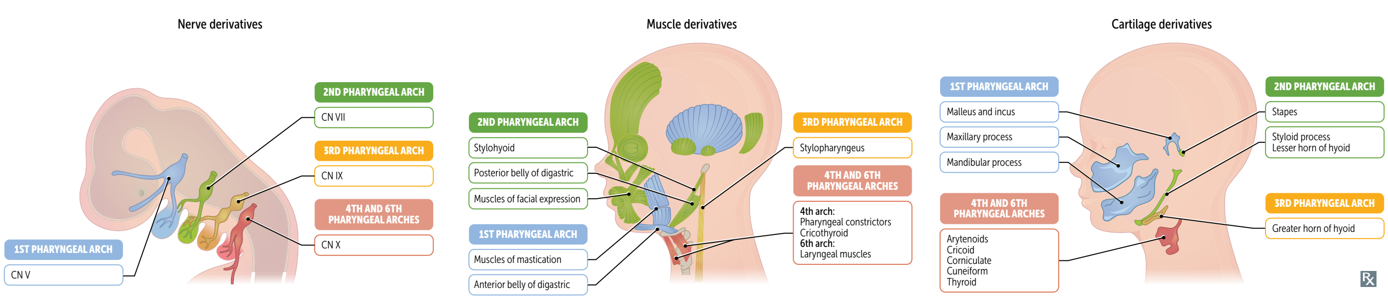
POUCH	DERIVATIVES	NOTES
1st pharyngeal pouch	Middle ear cavity, eustachian tube, mastoid air cells	1st pouch contributes to endoderm-lined structures of ear
2nd pharyngeal pouch	Epithelial lining of palatine tonsil	
3rd pharyngeal pouch	Dorsal wings → **inferior** parathyroids Ventral wings → thymus	**Thi**rd **p**ouch contributes to **th**ymus and both inferior **p**arathyroids Structures from 3rd pouch end up **below** those from 4th pouch
4th pharyngeal pouch	Dorsal wings → **superior** parathyroids Ventral wings → ultimopharyngeal body → parafollicular (C) cells of thyroid	**4**th pharyngeal pouch forms para"**4**"llicular cells Developmental anomalies in 3rd and 4th pharyngeal pouch are present in DiGeorge syndrome

Pharyngeal arch derivatives

Sensory and motor nerves are not pharyngeal arch derivatives. They grow into the arches and are derived from neural crest (sensory) and neuroectoderm (motor). Arches 3 and 4 form posterior 1/3 of tongue. Arch 5 makes no major developmental contributions.

When at the restaurant of the golden **arches**, children tend to first **chew** (1), then **smile** (2), then **swallow styl**ishly (3) or **simply swallow** (4), and then **speak** (6).

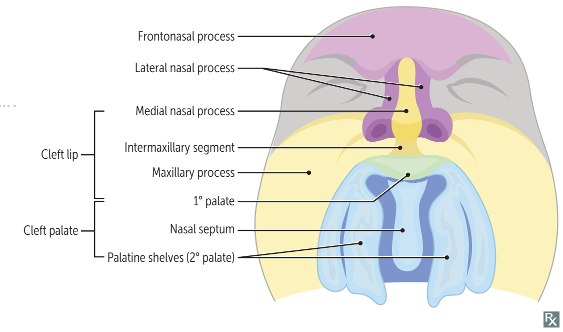
ARCH	NERVES	MUSCLES	CARTILAGE
1st pharyngeal arch	CN V_3 **chew**	Muscles of **m**astication (te**m**poralis, **m**asseter, lateral and **m**edial pterygoids), **m**ylohyoid, anterior belly of digastric, tensor tympani, anterior 2/3 of tongue, tensor veli palatini	**M**axillary process → **m**axilla, zygo**m**atic bone **M**andibular process → **M**eckel cartilage → **m**andible, **m**alleus and incus, spheno**m**andibular ligament
2nd pharyngeal arch	CN VII (**s**even)—**smile**	Muscles of facial expression, **s**tapedius, **s**tylohyoid, platy**s**ma, po**s**terior belly of digastric	Reichert cartilage → **s**tapes, **s**tyloid process, le**ss**er horn of hyoid, **s**tylohyoid ligament
3rd pharyngeal arch	CN IX—**swallow styl**ishly	**Styl**opharyngeus	Greater horn of hyoid
4th and 6th pharyngeal arches	4th arch: CN X (superior laryngeal branch) **simply swallow** 6th arch: CN X (recurrent laryngeal branch) **speak**	4th arch: most pharyngeal constrictors; cricothyroid, levator veli palatini 6th arch: all intrinsic muscles of larynx except cricothyroid	**A**rytenoids, **C**ricoid, **C**orniculate, **C**uneiform, **T**hyroid cartilage (used to sing and **ACCCT**)



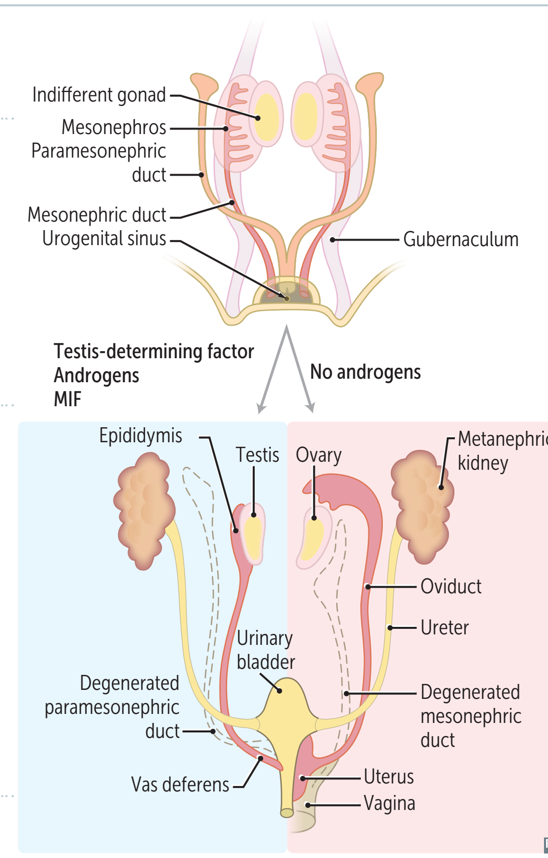
First and second pharyngeal arch syndromes

Pierre Robin sequence	Mandibular hypoplasia (micrognathia) → posteriorly displaced tongue (glossoptosis) → cleft palate, airway compromise. Feeding difficulties are common.
Treacher Collins syndrome	Autosomal dominant neural crest dysfunction → craniofacial abnormalities (eg, zygomatic bone and mandibular hypoplasia), hearing loss, airway compromise.

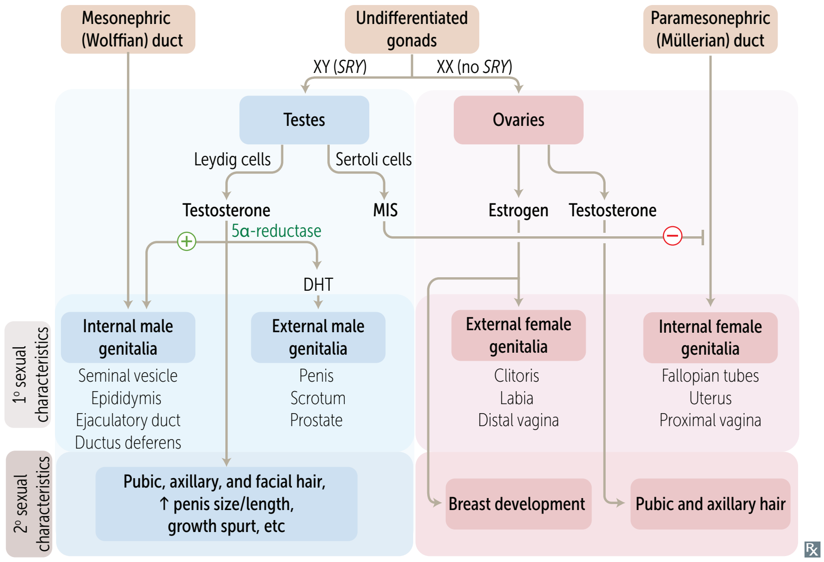
Orofacial clefts	Cleft lip and cleft palate have distinct, multifactorial etiologies, but often occur together.
Cleft lip	Due to failure of fusion of the intermaxillary segment (merged medial nasal processes) with the maxillary process (formation of 1° palate).
Cleft palate	Due to failure of fusion of the two lateral palatine shelves or failure of fusion of lateral palatine shelf with the nasal septum and/or 1° palate (formation of 2° palate).



Genital embryology	
Female	Default development. Mesonephric duct degenerates and paramesonephric duct develops.
Male	*SRY* gene on Y chromosome—produces testis-determining factor → testes development. Sertoli cells secrete Müllerian-inhibiting substance (MIS, also called anti-Müllerian hormone) that suppresses development of paramesonephric ducts. Leydig cells secrete androgens that stimulate development of mesonephric ducts.
Paramesonephric (Müllerian) duct	Develops into female internal structures—fallopian tubes, uterus, proximal vagina (distal vagina from urogenital sinus). Male remnant is appendix testis. Müllerian agenesis (Mayer-Rokitansky-Küster-Hauser syndrome)—1° amenorrhea with absent uterus, blind vaginal pouch, normal female external genitalia and 2° sexual characteristics (functional ovaries). Associated with urinary tract anomalies (eg, renal agenesis).
Mesonephric (Wolffian) duct	Develops into male internal structures (except prostate)—**S**eminal vesicles, **E**pididymis, **E**jaculatory duct, **D**uctus deferens (**SEED**). Female remnant is Gartner duct.



Sexual differentiation



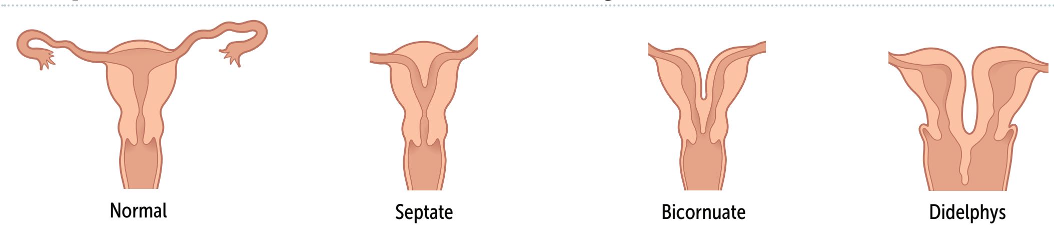
In XY individuals absence of Sertoli cells or lack of Müllerian-inhibiting substance → develop both male and female internal genitalia and male external genitalia (streak gonads)

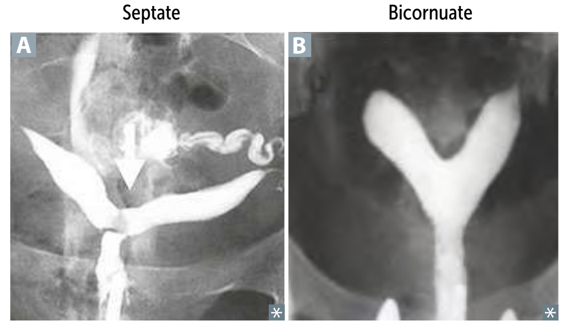
5α-reductase deficiency—inability to convert testosterone into DHT → male internal genitalia, atypical external genitalia until puberty (when ↑ testosterone levels cause masculinization)

In the testes:

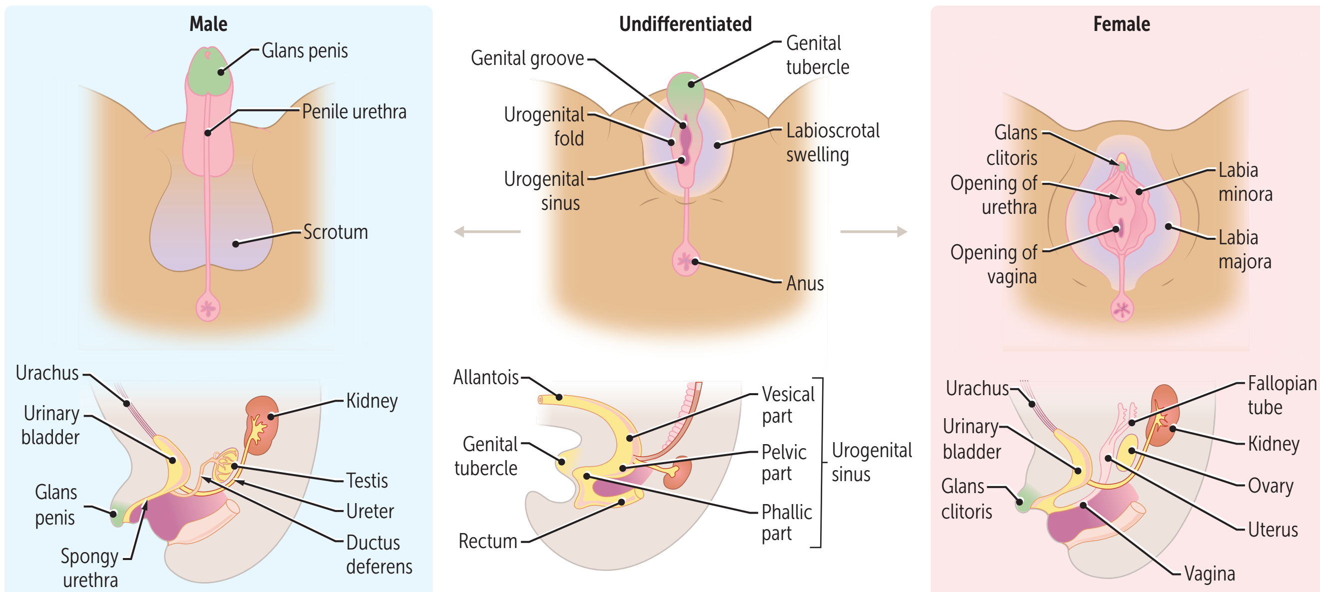
- Leydig leads to male (internal and external) sexual differentiation.
- Sertoli shuts down female (internal) sexual differentiation.

Uterine (Müllerian duct) anomalies	↓ fertility and ↑ risk of complicated pregnancy (eg, spontaneous abortion, prematurity, FGR, malpresentation). Hysterosalpingogram of normal uterus demonstrates normal uterine cavity and intraperitoneal spill of contrast (indicative of patent fallopian tubes).
Septate uterus	Incomplete resorption of septum A. Common anomaly. Treat with septoplasty.
Bicornuate uterus	Incomplete fusion of Müllerian ducts B.
Uterus didelphys	Complete failure of fusion → double uterus, cervix, vagina.



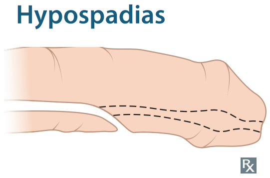
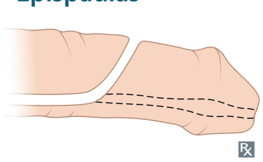


Male/female genital homologs



Dihydrotestosterone ←		→ Estrogen
Glans penis	Genital tubercle	Glans clitoris
Corpus cavernosum and spongiosum	Genital tubercle	Vestibular bulbs
Bulbourethral glands (of Cowper)	Urogenital sinus	Greater vestibular glands (of Bartholin)
Prostate gland	Urogenital sinus	Urethral and paraurethral glands (of Skene)
Ventral shaft of penis (penile urethra)	Urogenital folds	Labia minora
Scrotum	Labioscrotal swelling	Labia majora

Congenital penile abnormalities

Hypospadias 	Abnormal opening of penile urethra on ventral (**u**nder) surface due to failure of **u**rethral folds to fuse.	Hypospadias is more common than epispadias. Associated with inguinal hernia, cryptorchidism, chordee (downward or upward bending of penis). Can be seen in 5α-reductase deficiency.
Epispadias 	Abnormal opening of penile urethra on dorsal (**t**op) surface due to faulty positioning of genital **t**ubercle.	**E**xstrophy of the bladder is associated with **e**pispadias.

Descent of testes and ovaries

	DESCRIPTION	MALE REMNANT	FEMALE REMNANT
Gubernaculum	Band of fibrous tissue	Anchors testes within scrotum	Ovarian ligament + round ligament of uterus
Processus vaginalis	Evagination of peritoneum	Forms tunica vaginalis Persistent patent processus vaginalis → hydrocele	Obliterated

▶ REPRODUCTIVE—ANATOMY

Drainage of reproductive organs

Venous drainage

Right ovary/testis → right gonadal vein → IVC.
Left ovary/testis → left gonadal vein → left renal vein → IVC (takes the longer way).
Left testicular vein enters left renal vein at 90° angle → flow is less laminar on the left than on the right → left venous pressure > right venous pressure → varicocele is more common on the left.

Lymphatic drainage